

| Class      | Schema Data Type              | TLV Type                                                                   |
|------------|-------------------------------|----------------------------------------------------------------------------|
| Discrete   | enum8, enum16                 | Unsigned Integer (width is selected automatically depending on data value) |
|            | data8, data16, data32, data64 | Unsigned Integer (width is selected automatically depending on data value) |
|            | map8, map16, map32, map64     | Unsigned Integer (width is selected automatically depending on data value) |
|            | boolean                       | Boolean                                                                    |
| Composite  | string                        | UTF-8 string (length is selected automatically depending on data value)    |
|            | octstr                        | TLV octet string                                                           |
| Collection | list                          | TLV array                                                                  |
|            | struct                        | TLV structure                                                              |

### 10.3.1. Analog - Integer

All signed integer schema types SHALL be encoded using the TLV signed integer type. The specific TLV element type (1-byte, 2-byte, 4-byte and 8-byte signed integer types) SHALL be selected automatically at runtime depending on the actual value.

In this regard, the actual width of the over-the-wire type can be narrower than the width specified in schema.

E.g. a 32-bit value defined in schema will be encoded to a 1-byte TLV signed integer type if the value doesn't exceed (-128 to +127).

Similarly, all unsigned integer schema types SHALL be encoded using the TLV unsigned integer type.

### 10.3.2. Analog - Floating Point

Both single and double precision floating point analog schema types SHALL be encoded using equivalent TLV floating point types as well.

### 10.3.3. Discrete - Enumeration

Enumerations SHALL be encoded using the TLV unsigned integer type, with the width selected automatically at runtime based on the actual value.